NCBA AGENCY BANKING PROPOSAL

TANGAZOLETU LIMITED

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1. Executive Summary

This proposal outlines the development and implementation of an **Agency Banking System** for **NCBA Bank**, enabling cardless transactions through **APP** and **USSD** channels. The system will allow customers to perform various banking transactions (withdrawals, deposits, balance inquiries, etc.) without the need for physical cards, leveraging a network of agents and digital platforms. The solution will enhance customer convenience, reduce operational costs, and increase the bank's visibility and market reach.

The proposed system will include:

- **Agency Banking Portal** for agent Administration and monitoring agent performance.
- **Agent App and USSD** for facilitating transactions.
- **Integration with NCBA's existing systems** (NCBANOW app, USSD, T24).
- **Real-time transaction processing** to ensure instant updates and eliminate manual reconciliations.

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2. System Overview

2.1. Key Features

1. **Cardless Transactions**:
    - **Cardless Withdrawals**.
    - **Cardless Deposits**.
    - **Balance Inquiry**.
    - **Mini-Statement**.
    - **Third-Party Cash Withdrawals (Special Code)**.
2. **Multi-Channel Access**:
    - **Mobile App** (NCBANOW) for customers.
    - **USSD** for customers and agents without smartphones.
    - **Agent App** for transaction initiation and processing.
    - **Web-based Portal** for corporate and institutional agents (phase 2).
3. **Real-Time Integration**:

- Integration with **T24 Core Banking System** for real-time transaction processing.
- Integration with **NCBA Card Management System** for card-based transactions.
- **Open Banking APIs** for seamless transaction routing to T24.

4. **Security and Authentication**:
   - **One-Time Password (OTP)** for transaction verification.
   - **KYC Verification** by agents for cardless transactions.
   - **Geofencing and Geo tracking** of Agents using mobile app
   - **Agent working hours** enforcement for both USSD and Agent APP
   - **SIM Card binding**, IMSI and SIM Swap checks for Agents
   - **Fraud Detection** through real-time monitoring and reporting.
5. **Reporting and Monitoring**:
   - **Agent Performance Reports** (location, transaction volume, success rates).
   - **Transaction Reports** (successful, declined, fraud indicators).
   - **Reconciliation Reports** for failed transactions.
   - **Customizable Reports** in NCBA's desired format.

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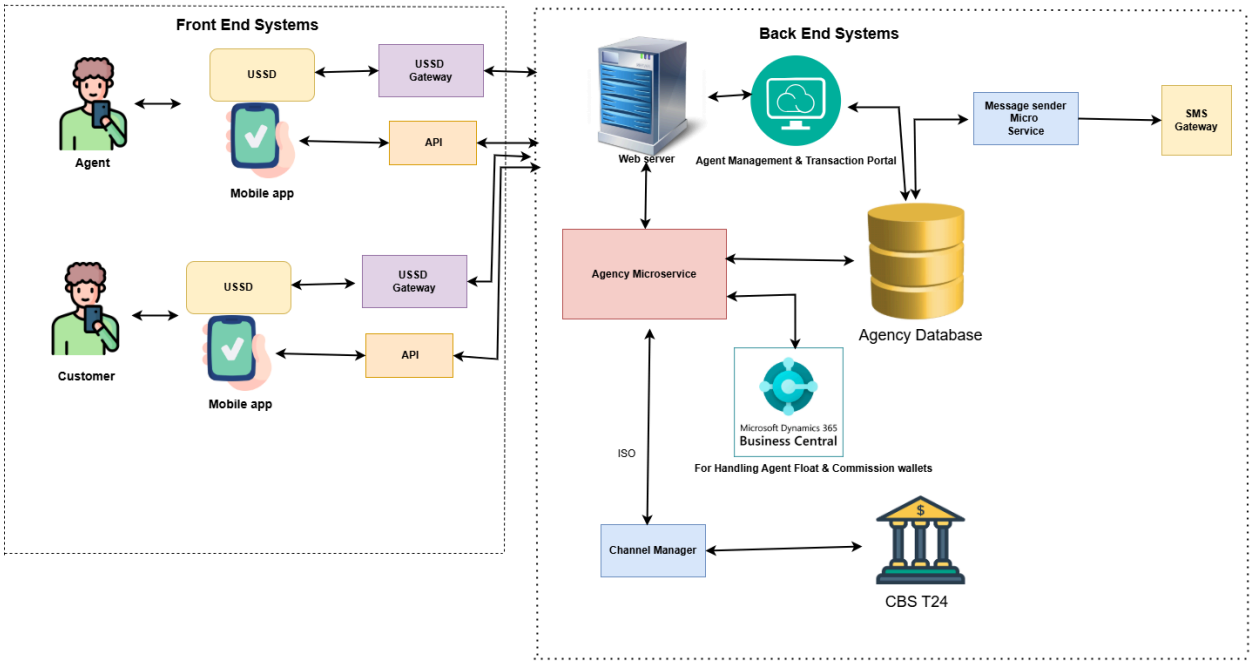
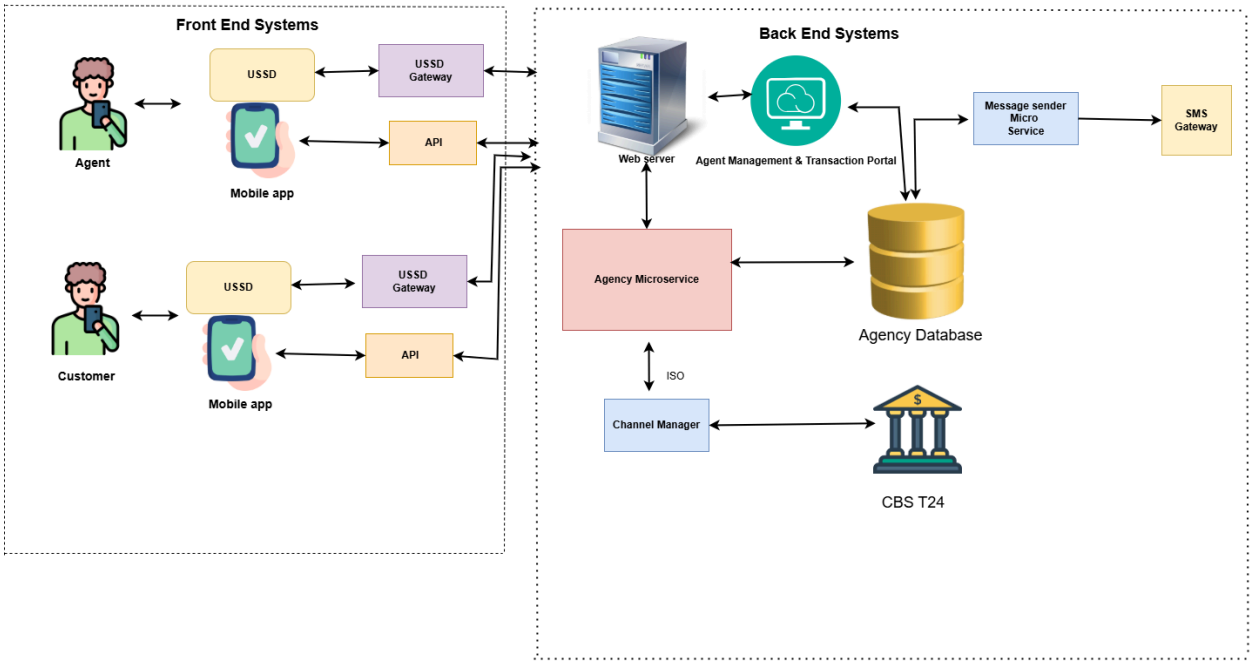
3. System Architecture

3.1. High-Level Architecture

Option 1 TL handling agents commissions and float accounts for Agents



Option 2 - Bank T24 handling agents commissions and float accounts



The proposed system will consist of the following components:

1. **Front-End Systems**:

- **Customer Channels**:
  - NCBA NOW Mobile App.
  - USSD (*488#).
- **Agent Channels**:
  - Agent Mobile App.
  - USSD for agents.
  - Web-based Portal (future phase).

2. **Back-End Systems**:
   - **Agency Banking Portal**:
     - Agent registration and management.
     - Transaction monitoring and reporting.
   - **Transaction Processing Engine**:
     - Integration with T24 for real-time transaction processing.
     - Integration with NCBA Card Management System for card-based transactions.
   - **Security and Authentication Layer**:
     - OTP generation and validation.
     - Geofencing and Geo tracking of Agents using mobile app
     - Agent working hours enforcement for both USSD and Agent APP
     - SIM Card binding, IMSI and SIM Swap checks for Agents
     - Fraud detection and prevention.
3. **Integration Layer**:

- **Open Banking APIs** for seamless integration with T24 and any other necessary bank system.
- **ELMA System** for transaction validation and routing to the agency system.

4. **Database Layer**:
   - **Agent Database**: Stores agent details, transaction history, and performance metrics.
   - **Transaction Database**: Stores all transaction records for reporting and reconciliation.

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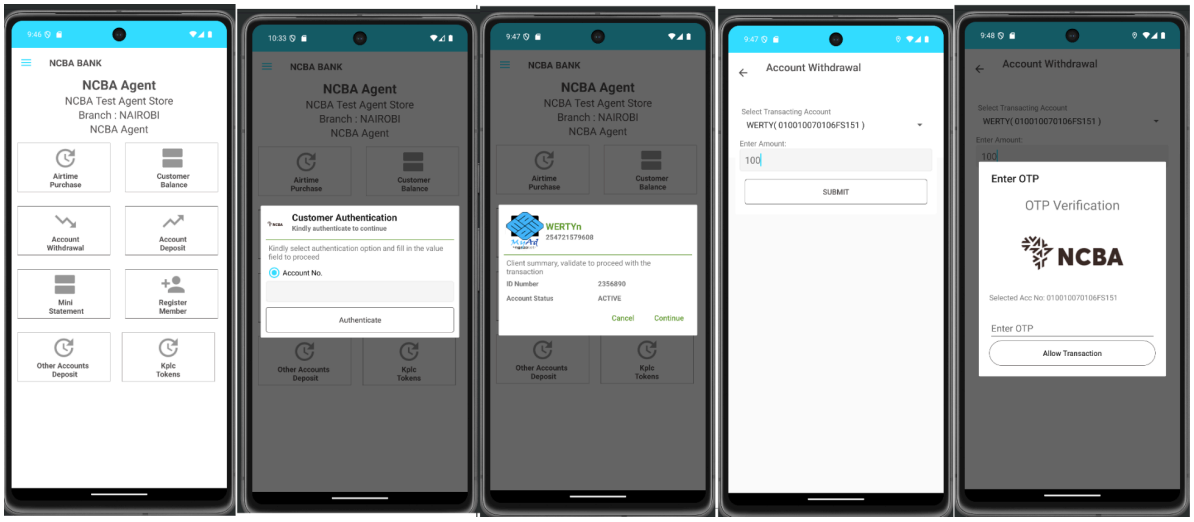
4. Functional Requirements

4.1. Cardless Withdrawal (Agent)

- **Customer visits the agent and provides** account details.
- **Agent keys in details** on the POS, and customer confirms.
- **OTP generated** and sent to the customer's registered phone number.
- **Agent verifies OTP**, by keying in for verification and authorization of transaction.
- **Transaction is** executed in T24 via Open Banking APIs integration.
- **Receipt generated**, and cash issued to the customer.

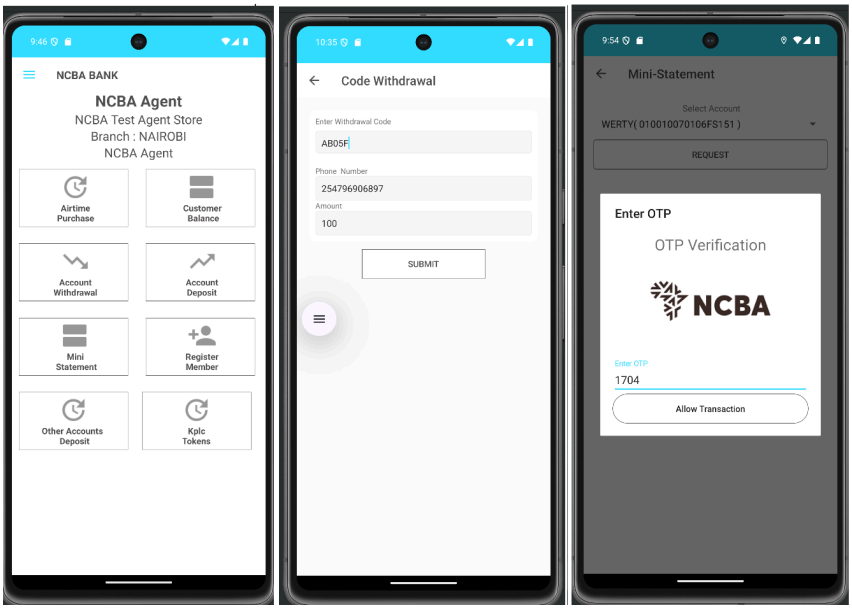


4.2. Cardless Withdrawal (Third Party)

- **Customer initiates withdrawal** for a third party via NCBANOW app or USSD.
- **Special Withdrawal Code is generated** and sent to the recipient's phone number.
- **Recipient visits agent** and provides a Special Code, amount and phone number.

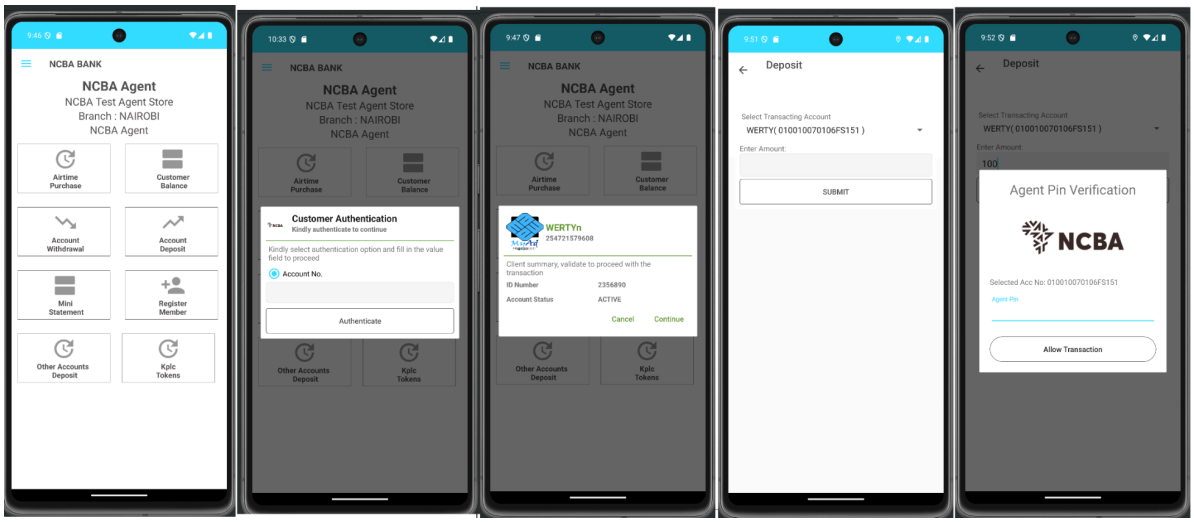
- **Transaction authorized and executed** by T24 via Open Banking APIs integration.
- **Cash issued** to the recipient, and confirmation SMS sent to both parties.

Customer Initiates the Third Party Withdrawal from either NCBA NOW or USSD *488#



4.3. Cardless Deposit

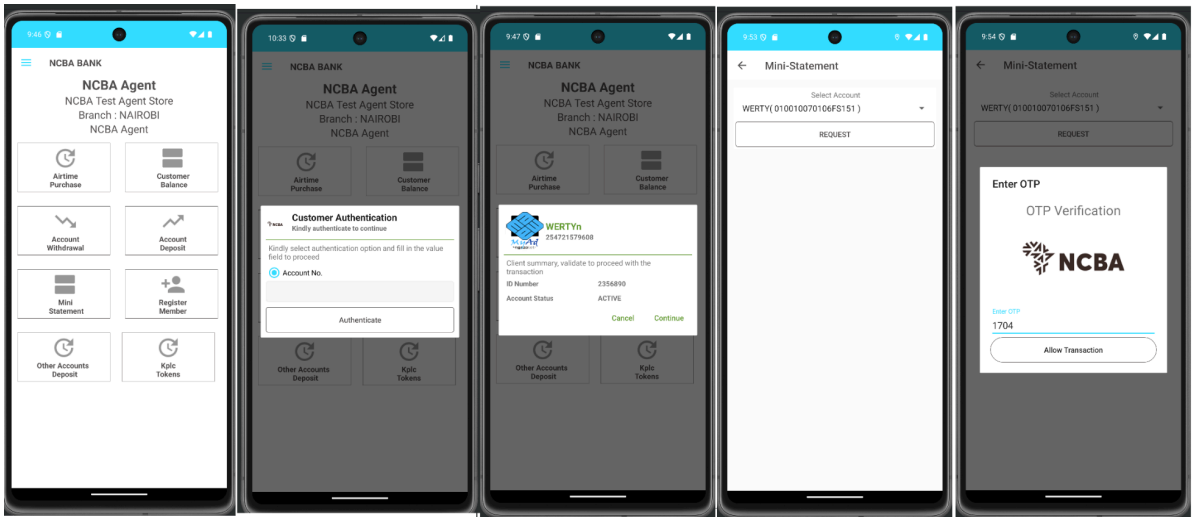
- **Customer visits the agent** and provides account details and cash.
- **Agent keys in details** on the POS, and customer confirms.
- **Transaction routed** to T24 via Open Banking APIs.
- **Customer account credited** in real-time, and confirmation SMS sent.



4.4. Balance Inquiry and Mini-Statement

- **Customer visits agent**, provides account details and ID.

- **Agent selects transaction type** on the POS.
- **OTP is** generated and sent to the customer
- **Agent keys in the OTP** the same is verified, and transaction is processed.
- **Balance or mini statement** printed and handed to the customer.



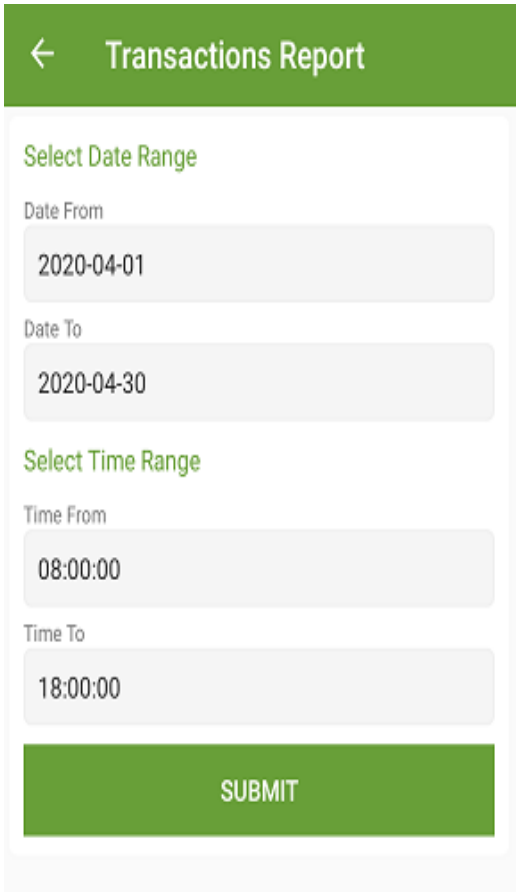
4.5 Agent Transaction History Report

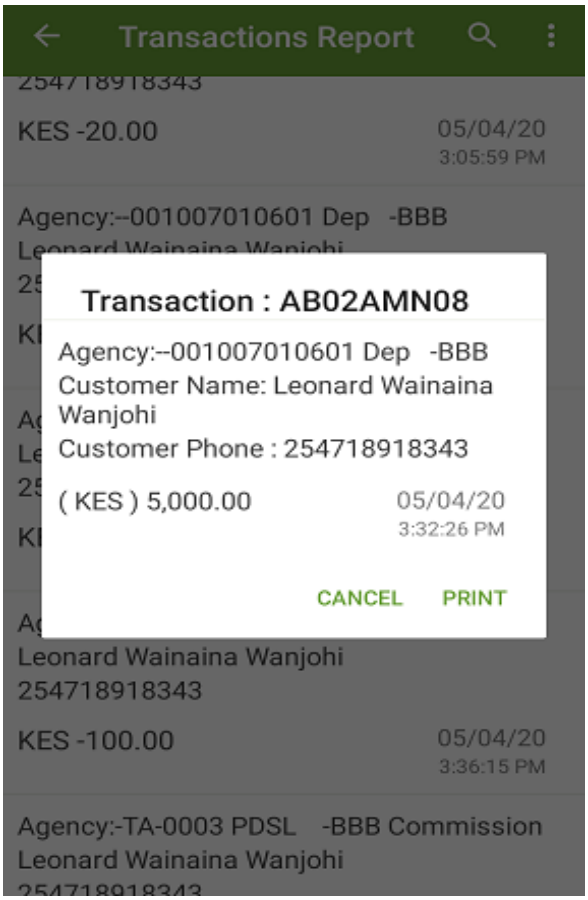
The Agent Transaction History Report also referred to as the end of day report helps the agent as well as the Organization track transactional activities for the registered Agent.

The report can be pulled for as far back as 3 months and enables the agent to print it out as shown on below steps and captures.

Capture: Z-report

Capture 6.3: Printing





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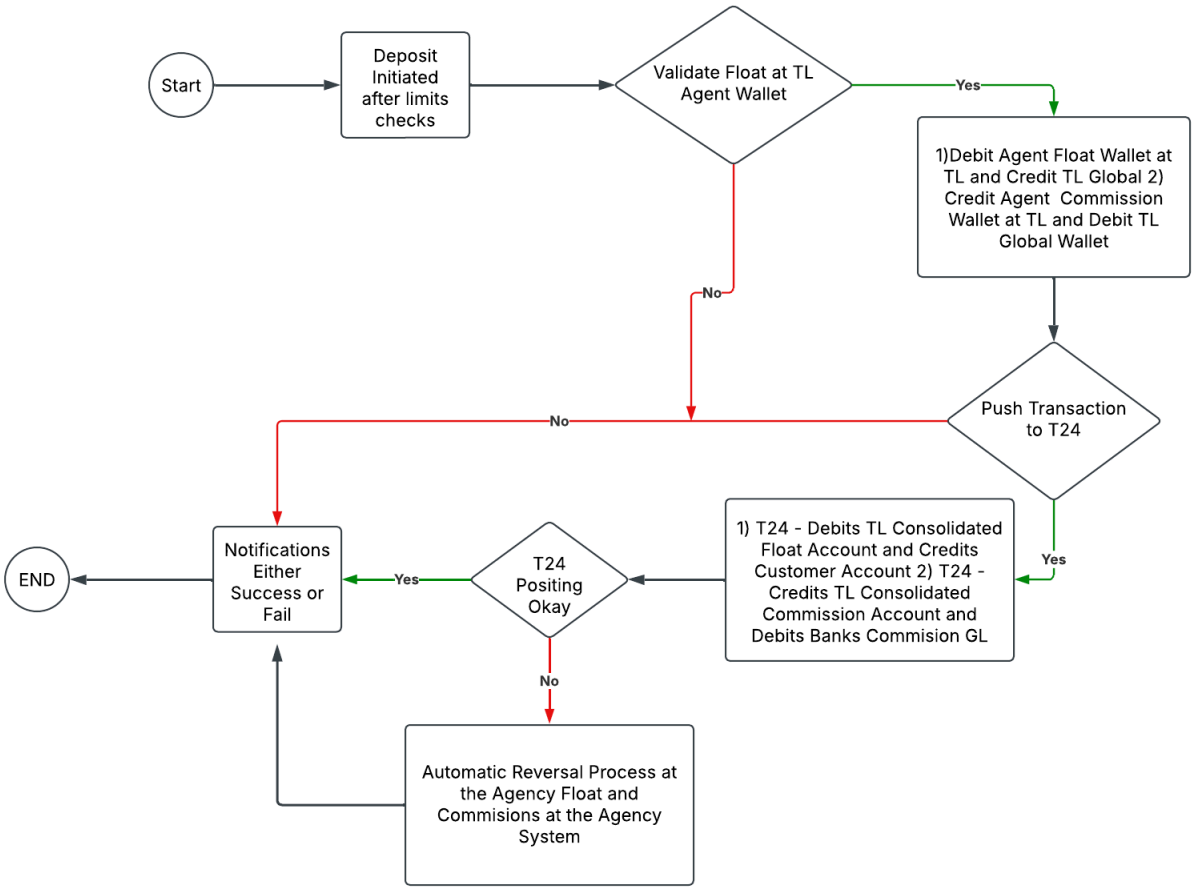
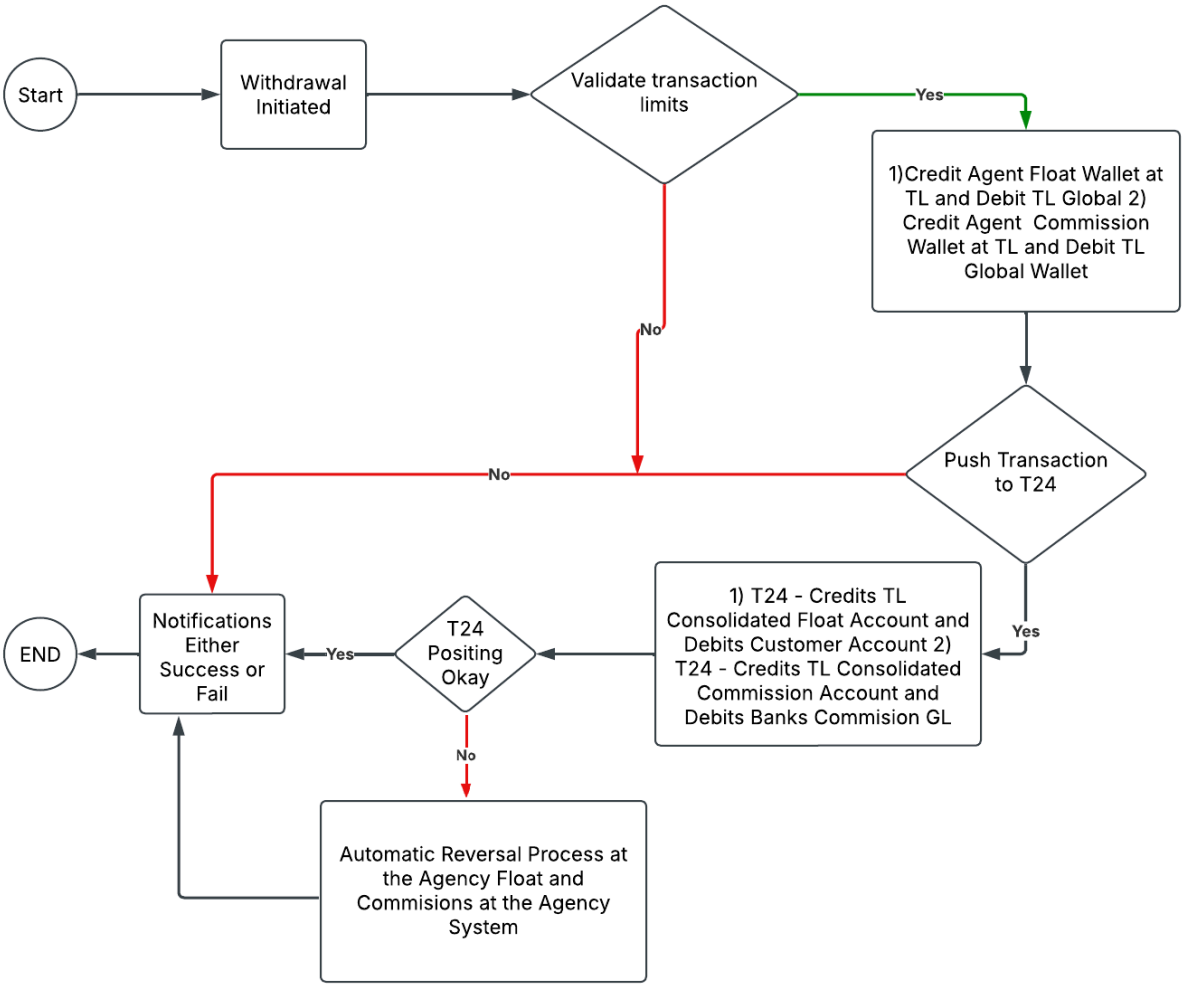
5. Transactions Process Flows

5.1 Tangazoletu handling Agents Wallets and T24 having TL Consolidated Accounts for Float and Commissions

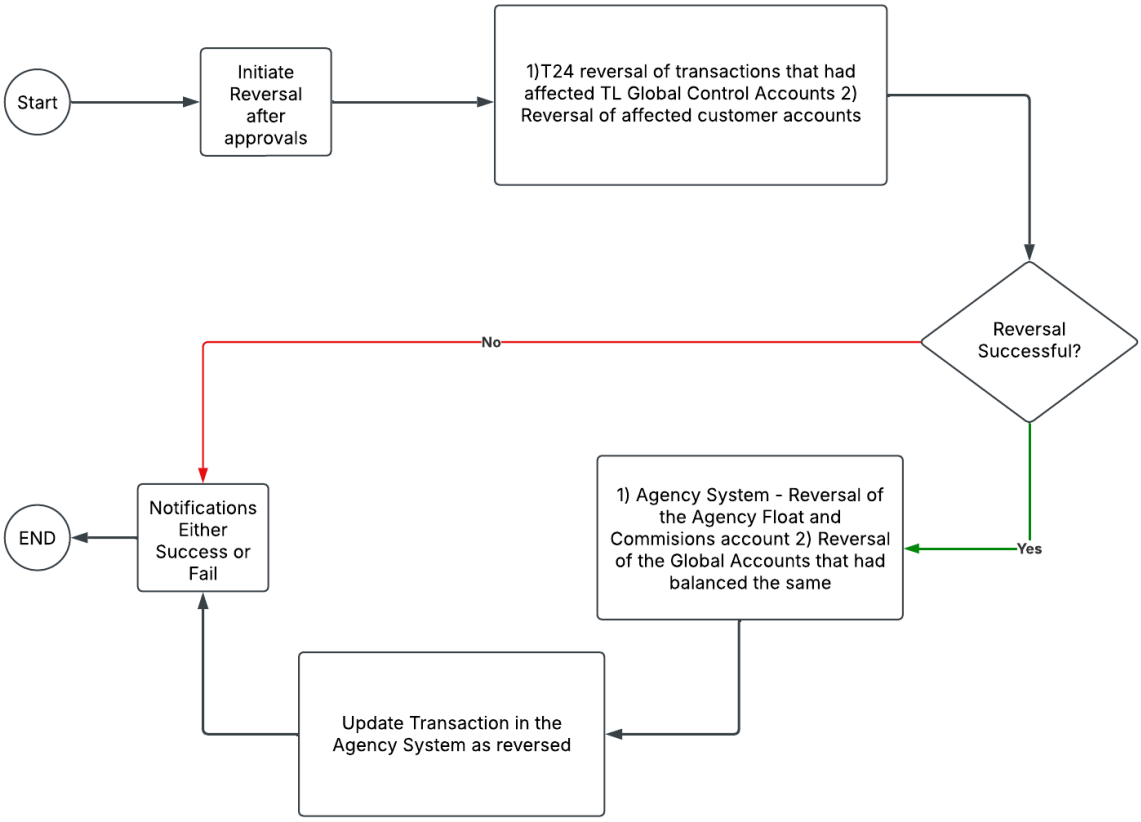
a) **Deposit**



b) Withdrawal

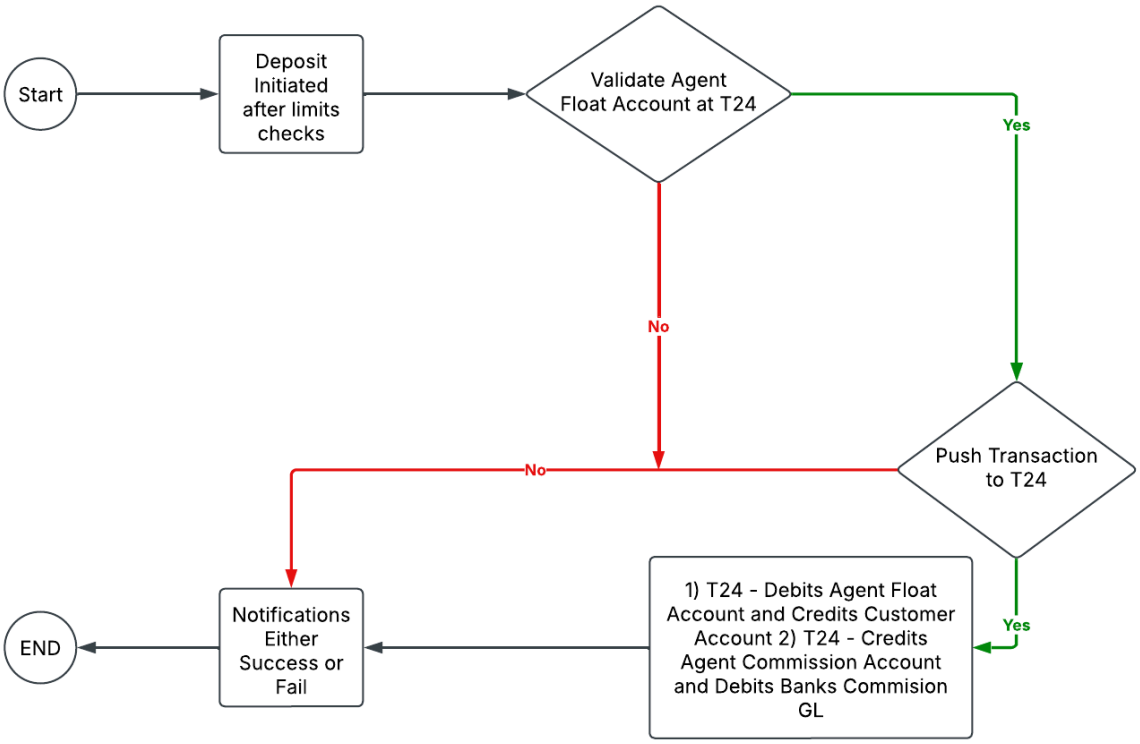


c) Manual Reversal

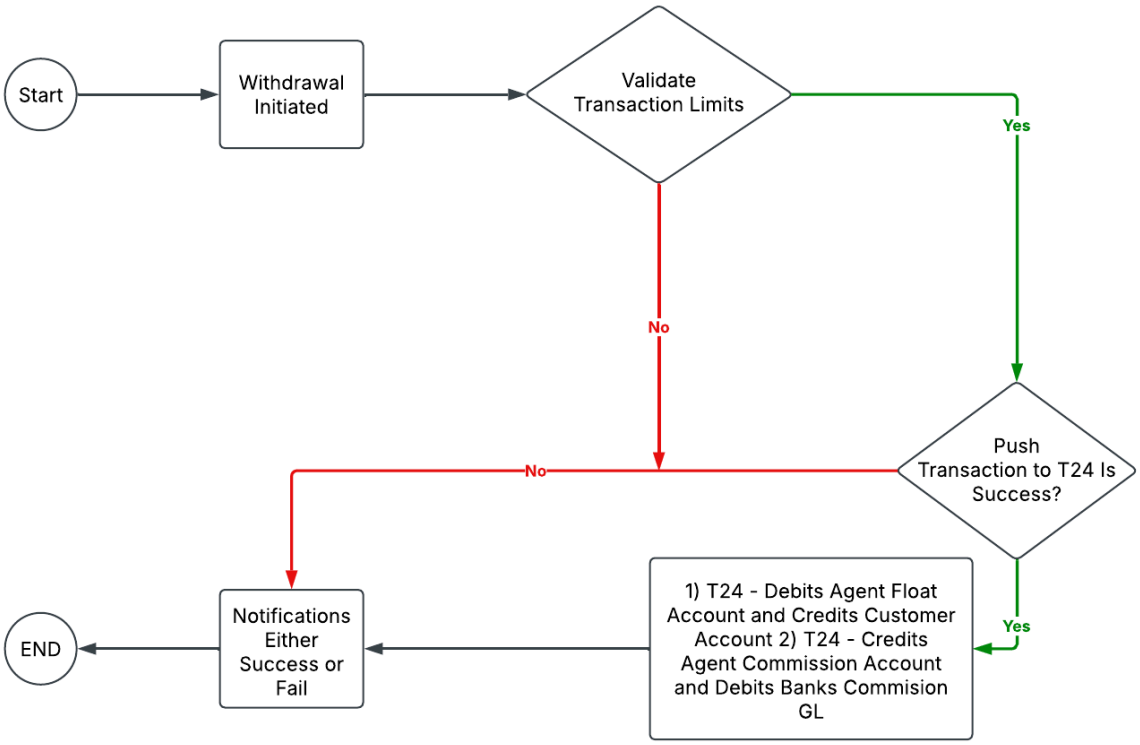


5.2 T24 handling Agents Accounts for Float and Commissions

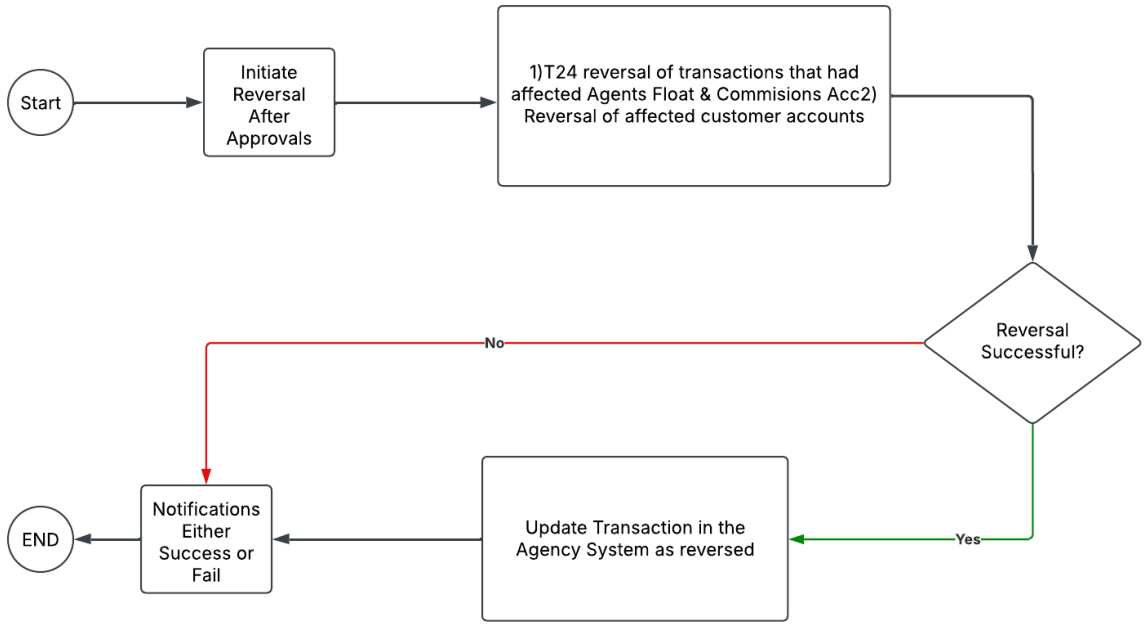
d) Deposit



e) Withdrawal



f) manual reversal



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6. Non-Functional Requirements

6.1. Reporting

- **Agent Performance Reports**: Agent number, location, transaction volume, success rates.
- **Transaction Reports**: Successful, declined, and fraud indicators.
- **Reconciliation Reports**: For failed transactions.
- **Customizable Reports**: In NCBA's desired format.

6.2. Security

- **OTP Authentication**: For all cardless transactions.
- **Geofencing and Geo tracking** of Agents using mobile app
- **Agent working hours** enforcement for both USSD and Agent APP
- **SIM Card binding**, IMSI and SIM Swap checks for Agents
- **Fraud Detection**: Real-time monitoring and reporting.
- **KYC Verification**: By agents for cardless transactions.

6.3. Performance

- **Real-Time Processing**: Transactions routed through T24 in real-time.
- **System Downtime Handling**: Transactions declined during downtime to prevent errors.

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7. Implementation Plan

7.1. Phase 1: Development and Testing

- **Duration**: 1.5 months.
- **Activities**:
  - Configure Agents Management Portal.
  - Configure Agent App and USSD.
  - Integrate with T24 and NCBA Card Management System.
  - Integrations with NCBA NOW and Customer USSD
  - Conduct end-to-end testing.

7.2. Phase 2: Pilot Launch

- **Duration**: 2 weeks.
- **Activities**:
  - Pilot launch with a select group of agents.
  - Monitor system performance and gather feedback.
  - Make necessary adjustments.

7.3. Phase 3: Full Deployment

- **Duration**: 1 month.
- **Activities**:
  - Full deployment across all agents.
  - Train agents and customers on the new system.
  - Monitor system performance and provide ongoing support.

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8. Support

TANGAZOLETU LIMITED SLA (SERVICE-LEVEL AGREEMENT)
PRIORITY MATRIX and Response Targets

Impact and Urgency

1. Critical: Priority 1 (P1)

Critical production issue that severely impacts the use of the service. The situation halts the business operations, and no procedural workaround is immediately available. Service is down or unavailable.

1. Components –P1

- ✔ Data corrupted or lost and must restore from backup.
- ✔ A critical documented feature / function is not available.
- ✔ All or a substantial portion of critical data is at a significant risk of loss or corruption.
- ✔ Substantial loss of service.
- ✔ Business operations have been severely disrupted.
- ✔ P1 issues require the customer to have dedicated resources available to work on the issue on an ongoing basis.

2. Major: Priority 2 - (P2)

Major functionality is impacted, or significant performance degradation is experienced. The situation is causing a high impact to portions of the business operations.

Major functionality is severely impaired.

1. Components –P2

- ✔ Service is operational but highly degraded performance to the point of major impact on usage. Important features of the Software as a Service offering; however, operations can continue in a restricted fashion although the long-term productivity might be adversely affected. A major milestone is at risk. Ongoing and incremental installations are affected.
- ✔ A temporary workaround is available.

3. Minor: Priority 3 - (P3)

There is a partial, non-critical loss of use of the service with a medium-to-low impact on your business, but your business continues to function. Short-term workaround is available, but not scalable.

1. Components –P3

- ✔ Partial, non-critical loss of functionality of the software.
- ✔ Impaired operations of some components but allows the user to continue using the software.

- ✔ Initial installation milestones are at minimal risk.

4. General usage questions.

Cosmetic issues, including errors in the documentation.
Depending on the impact and urgency, a major incident will be categorized as a P1 or P2. All P1 tickets are considered major incidents. P2 tickets are considered major if the impact is “multiple groups”

P1 major incidents are worked 24/7. P2 major incidents are worked until completed, including after hours.

1. Components –P4

- ✔ Inquiry regarding a routine technical issue; information requested on application capabilities, navigation, installation, or configuration; bug affecting a small number of users. Acceptable workaround available.